



DPUSGECapture

This DPU can be used to create digital videos during the simulation. The video contains graphics from a SGEView window, or alternatively the graphics of a DPUBirdsEye or DPUMMI window. Optionally, audio can be captured from a sound card. Using the DPUSGETexture DPUs, external image sources such as cameras can be displayed in a SGEView window and thus DPUSGECapture can also be used to capture these images.

The video file format can be chosen freely, it is selected by the extension of the output file (DPU parameter *FileName*).

By default, standard video and audio codecs are selected matching the output file, but optionally other codecs can be selected (DPU parameter *VideoCodec*). The video file is created using the library libavcodec by the FFMpeg project (<http://www.ffmpeg.org>), which supports many common video and audio formats.

1. Static parameters:

<i>Source</i>	DPU instance name of the DPU whose graphics output will be captured. This can be an instance of DPUSGEView, DPUBirdsEye or DPUMMI. The default value is "SGEView", which is the instance name of the SGE view that is defined in SILAB's default configuration.
<i>Subsample</i>	Proper divisor of the frame rate. The following equation is valid: $\text{Video frame rate} = \text{SGEView update rate} / \text{Subsample}$ Default: 1
<i>Path</i>	Output path of recorded file. If no absolute path is provided the path starts in the installation path of SILAB. Default: "Data\"
<i>FileName</i>	File name of the created video (within the Path folder). Default: "SGECapture" Alternatively, it can be specified using the launch data <code>VAR_DATA-FILENAME</code> .
<i>Suffix</i>	Suffix that is inserted into the file name (after <i>FileName</i> , but before <i>Extension</i>). Default: (none)
<i>Extension</i>	File name extension (without leading dot). Default: "mp4". It is only appended to the file name if <i>FileName</i> doesn't already contain an extension.
<i>Overwrite</i>	If set to <i>true</i> , an already existing video file of the given file name is overwritten.



	If set to <i>false</i> (the default), numbered video files are created instead.
<i>DiskFree_Warning_MB</i>	Minimum recommended free disk space (MB) before starting a simulation. If less space is free, a warning is issued. Default: 10000
<i>DiskFree_Error_MB</i>	Minimum required free disk space (MB) before starting a simulation. If less space is free, the simulation cannot be started, because the space will probably run out. Default: 500

2. Static parameters concerning video compression:

<i>VideoCodec</i>	Name of the video codec that is used for compression. Default: empty value (<i>VideoCodec</i> = ""), which means that a suitable video codec is chosen automatically. The section "Functional principle" contains example configurations for some of the available codecs.
<i>VideoOptions</i>	Additional options for the <i>VideoCodec</i> . The options are specified in the format " <i>key1=val1 key2=val2 ...</i> ". These options are set at the stream format level.
<i>VideoCodecOptions</i>	Additional options for the <i>VideoCodec</i> . The options are specified in the format " <i>key1=val1 key2=val2 ...</i> ". These options are set at the encoder level.
<i>VideoKbps</i>	Data rate (in kbit/s) of the created video stream. Depending on the video resolution and codec, values between 1000 (1 Mbps) and 25000 (25 Mbps) should be used. Default: 3000
<i>HighVideoQuality</i>	If true is entered, the video codec works in its high quality setting. That means that the same bit rate will yield better results, but the codec will consume more CPU time. Please check the output <i>NDroppedFramesTotal</i> to determine if <i>HighVideoQuality</i> can be used. Default: <i>false</i>
<i>Threads</i>	Number of encoder threads. The number should match the number of available CPU cores. Default: 2 Warning: Not all video codecs support parallel encoding, in that case a warning message will appear in SILAB and <i>Threads=1</i> must be entered.

3. Static parameters concerning audio capture:

<i>CaptureAudio</i>	Should audio be captured from an external source (such as the Mic-In input of the computer's sound card)? Default: <i>false</i> (output video will be silent)
<i>AudioDevice</i>	Name of the audio device to capture from. Default: (<i>empty</i>), i.e. the default device is used
<i>AudioKbps</i>	Data rate (in kbit/s) of the created audio stream. Depending on the audio codec, values between 64 and 500 kbps should be used. Default: 128
<i>AudioSampleRate</i>	Sample rate used when capturing from an external source.



MaxAudioSkewMS Default: 44100 (this rarely needs to be changed)
Controls audio synchronization. This is the max. allowed skew between the audio and video streams before audio is resynchronized.
Default: 50

4. Outputs:

VideoTimeMS Current video time stamp, in milliseconds. If a DPUDataFile recording is running in parallel, it is recommended to add this variable to the DPUDataFile, so that the files can be synchronized precisely afterwards.

NDroppedFramesCurrent Counts the number of video frames that had to be dropped within the last second due to CPU overload.

NDroppedFramesTotal Counts the number of video frames that had to be dropped during the whole simulation due to CPU overload. This value should always be less than five; otherwise some parameters must be reduced.

FrameNumber Current frame number within the created video. This value can also be used to synchronize the video later to data files recorded in parallel.

EncodeAvgMS Specifies the average time (ms) required to encode a single video frame. This value can be used to check how much CPU time is still available.

5. Launchdata:

VAR_DATAFILENAME Overrides the FileName that was specified in the configuration.

6. Dependencies to other DPUs:

DPUSGECapture must run together with the graphics DPU whose image is recorded. The *Index* value of DPUSGECapture must be higher than the *Index* value of the graphics DPU.

7. Recommendation:

- Computing the simulation's 3D graphics and encoding the video at the same time imposes high demands on the computer. A quad-core processor is recommended.

8. Functional principle:

The DPU works well when using the following script:

```
DPUSGECapture SGECap
{
    Index = 100;
    Computer = {REC,LOCALHOST};

    Subsample = 2;

    /* # recommended settings for MPEG-4 AVC video hardware encoder,
    # which can be used only if a recent NVidia card is present
    # (GeForce GTX 650 or higher).
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
# The hardware encoder provides better quality at lower file sizes
# and lower CPU load than the software encoders listed below.
FileName = "SGECaptureTest4.mp4"; # MPEG4 AVC video, AAC audio
VideoCodec = "h264_nvenc"; # select NVidia hardware encoder
VideoKbps = 8000;
*/
/* recommended settings for MPEG-4 video, using the software en-
coder:
FileName = "SGECaptureTest4.mp4"; # MPEG4 video, AAC audio
VideoKbps = 8000;
*/
/* recommended settings for MPEG-2 video:
FileName = "SGECaptureTest2.mpg"; # MPEG2 video, MP2 audio
VideoKbps = 8000;
*/
/* recommended settings for MPEG-4 AVI video:
FileName = "SGECaptureTest4.avi"; # MPEG4 in AVI codec, ADPCM audio
VideoOptions = "bf=0"; # disabling B-frames improves compatibility
VideoKbps = 8000;
*/
/* recommended settings for MJPEG video:
FileName = "SGECaptureTest4.avi"; # MJPEG in AVI file, ADPCM audio
VideoCodec = "mjpeg";
VideoKbps = 42000;
*/
};

REC.SGEView.VerticalSync = false;
REC.SGEView.WindowW = 960;
REC.SGEView.WindowH = 600;
```